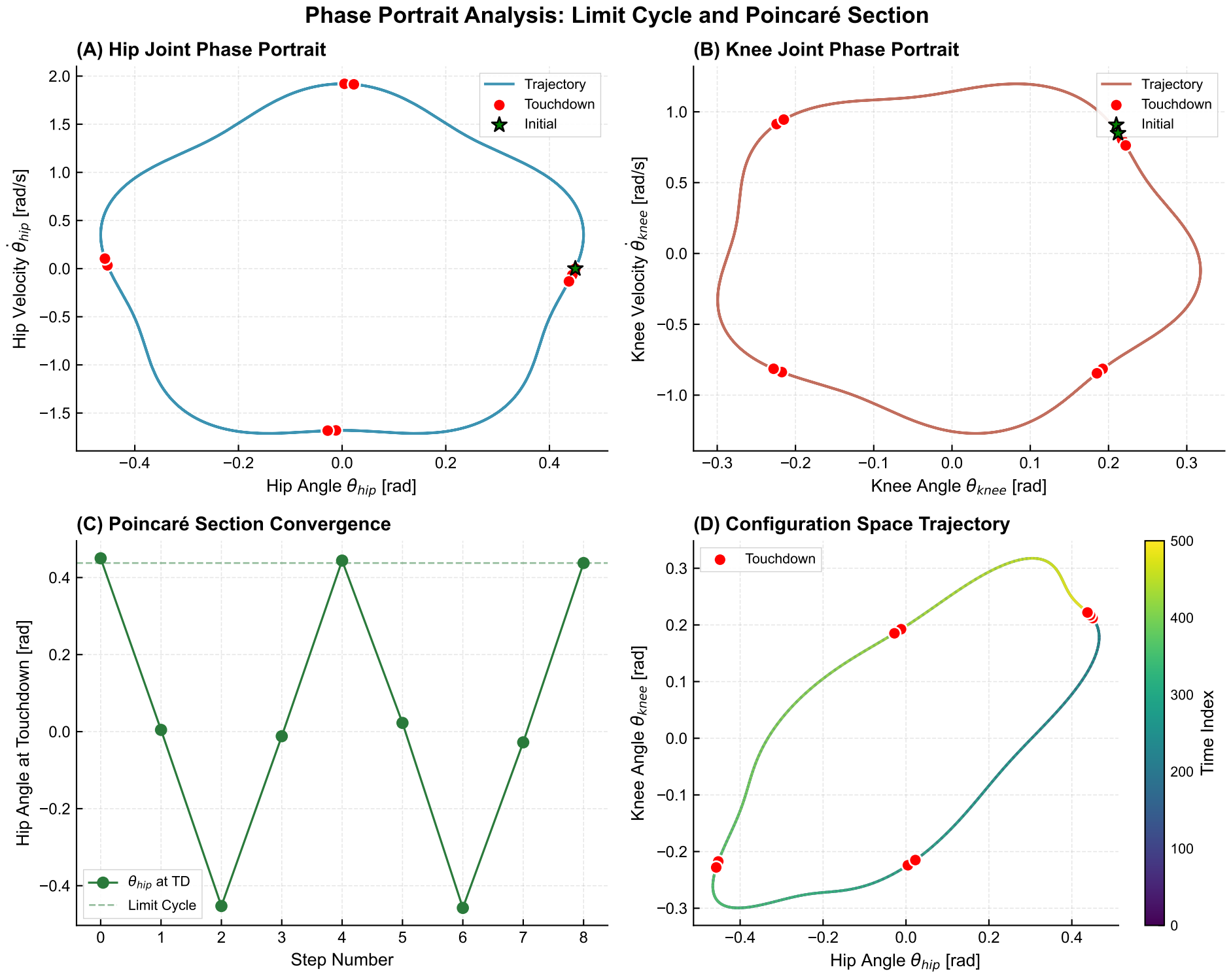

Phase Portrait Analysis: Limit Cycle and Poincaré Section
(A) Hip Joint Phase Portrait
Trajectory
Touchdown
Initial
Hip Velocity $\dot{\theta}_{hip}$ [rad/s]
Hip Angle $\theta_{hip}$ [rad]
(B) Knee Joint Phase Portrait
Trajectory
Touchdown
Initial
Knee Velocity $\dot{\theta}_{knee}$ [rad/s]
Knee Angle $\theta_{knee}$ [rad]
(C) Poincaré Section Convergence
$\theta_{hip}$ at TD
Limit Cycle
Hip Angle at Touchdown [rad]
Step Number
(D) Configuration Space Trajectory
Touchdown
Knee Angle $\theta_{knee}$ [rad]
Hip Angle $\theta_{hip}$ [rad]
Time Index